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Inventor(s)

Watts; Michael Robert et al.

MANAGING MODULE INTERFACES FOR MODULAR OPTICAL SYSTEMS

Abstract

A first (second) chip comprises a first (second) surface in proximity to a first (second) set of two or more waveguide couplers. The chips are optically coupled to each other by an interface between the surfaces in which: a first waveguide coupler of the first set is in proximity to a second waveguide coupler of the second set within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, and a third waveguide coupler of the first set is in proximity to a fourth waveguide coupler of the second set within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region.

Inventors: Watts; Michael Robert (Hingham, MA), Guglielmon; Jonathan Andrew (Cambridge, MA), Xie; Shengjie (Brighton, MA), Moss; Benjamin Roy (Cambridge, MA), Nickerson; Michael J. (Boston, MA)

Applicant: Analog Photonics LLC (Boston, MA)

Family ID: 96881109

Assignee: Analog Photonics LLC (Boston, MA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of and priority to U.S. Provisional Application Ser. No. 63/559,417, entitled “MANAGING MODULE INTERFACES FOR MODULAR OPTICAL SYSTEMS,” filed Feb. 29, 2024, which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates to managing module interfaces for modular optical systems.

BACKGROUND

[0003] Chip-scale devices and systems comprising integrated circuits have applications ranging from electronics to optical connectivity. Some integrated circuit devices are configured to guide or manipulate optical waves using photonic components or subcomponents. Such integrated circuit devices can be referred to as photonic integrated circuits or PICs. Some PICS comprise “active” components that are configured to receive power, such as a voltage or current, from external circuitry. Some PICs comprise “passive” components that are configured to guide or direct optical waves or photonic signals without power from external circuitry. Some PICs are configured as passive photonics chips such that at least a portion of the passive photonics chip comprises components that do not use power from external circuitry.

[0004] In some examples, PICs comprising active and passive components can be used for light detection and ranging (LiDAR) systems, where an optical wave from an optical source can be transmitted to target object(s) at a given distance and the light backscattered from the target object(s) can be collected. Various techniques, such as modulation and/or time of flight, can be used to determine distance to a target object based on information associated with a detection event.

[0005] The functionality of some photonic microsystems or PICs can be sensitive to the precise performance of very specific subcomponents. In many cases, suboptimal performance of a subcomponent is tolerated to facilitate integration of the subcomponents into a unified fabrication process flow with other components. For example, the loss of nitride waveguides can be improved through high-temperature annealing; however, annealing may be incompatible with doped active components, so often higher nitride waveguide loss is tolerated to facilitate its inclusion in an active process.

[0006] The optical source used in such photonic systems can be a laser or other coherent light source, which enables photonic signals to be carried on an optical wave that has as narrow linewidth and has a peak wavelength that falls in a particular range (e.g., between about 100 nm to about 1 mm, or some subrange thereof), also referred to herein as simply “light.”

SUMMARY

[0007] In one aspect, in general, an apparatus comprises: a first chip comprising a first surface in proximity to a first set of two or more waveguide couplers; and a second chip comprising a second surface in proximity to a second set of two or more waveguide couplers; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more waveguide

couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region, and the first coupling region is separated from the second coupling region by a separation distance that is larger than a maximum length of the first coupling region and larger than a maximum length of the second coupling region.

[0008] Aspects can include one or more of the following features.

[0009] Each waveguide coupler in the first set of two or more waveguide couplers is at an end of a corresponding waveguide on first chip, and each waveguide coupler in the second set of two or more waveguide couplers is at an end of a corresponding waveguide on the second chip.

[0010] The corresponding waveguides on the first chip are arranged (1) to have propagation axes that are all parallel to a common axis and (2) such that corresponding waveguide couplers of waveguides that are adjacent to each other on the first chip are offset from each other along the common axis.

[0011] The first waveguide coupler and the third waveguide coupler are substantially coplanar with a first plane that is parallel to the common axis and the second waveguide coupler and the fourth waveguide coupler are substantially coplanar with a second plane that is parallel to each of the common axis and the first plane.

[0012] A portion of the first waveguide coupler and a portion of the second waveguide coupler within the first coupling region are contained within a third plane that is perpendicular to the first plane and perpendicular to the common axis, and a portion of the third waveguide coupler and a portion of the fourth waveguide coupler within the second coupling region are contained within a fourth plane that is perpendicular to the first plane and perpendicular to the common axis.

[0013] The third plane and the fourth plane are distributed along the common axis.

[0014] Each of the first waveguide coupler, the second waveguide coupler, the third waveguide coupler, and the fourth waveguide coupler comprise respective grating couplers.

[0015] Each waveguide coupler of the first set of two or more waveguide couplers is optically coupled to a different respective optical phase shifter.

[0016] Each waveguide coupler of the second set of two or more waveguide couplers is optically coupled to a different respective optical antenna.

[0017] The first surface of the first chip is in proximity to a third set of one or more waveguide couplers and the second surface of the second chip is in proximity to a fourth set of one or more waveguide couplers.

[0018] The first chip and the second chip are optically coupled to each other by a second interface between the first surface and the second surface in which: a fifth waveguide coupler of the third set of one or more waveguide couplers is in proximity to a sixth waveguide coupler of the fourth set of one or more waveguide couplers within a distance that supports evanescent optical coupling between the fifth waveguide coupler and the sixth waveguide coupler that are overlapped within a third coupling region.

[0019] At least one waveguide coupler in the fourth set of one or more waveguide couplers is in optical communication with an optical source on the second chip.

[0020] In another aspect, in general, an apparatus comprises: a first chip comprising a first surface in proximity to a first set of two or more waveguide couplers; and a second chip comprising a second surface in proximity to a second set of two or more waveguide couplers; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more

waveguide couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region, the first waveguide coupler and the third waveguide coupler are each at an end of a waveguide that is coupled to an active photonic component, and the second waveguide coupler and the fourth waveguide coupler are each at an end of a waveguide that is coupled to a passive photonic component.

[0021] Aspects can include one or more of the following features.

[0022] The active photonic component comprises an optical phase shifter.

[0023] The first chip comprises one or more through oxide vias in electrical communication with the active photonic component.

[0024] The passive photonic component comprises an optical antenna.

[0025] Each of the first waveguide coupler and the second waveguide coupler comprise a respective adiabatic taper within the first coupling region, and each of the third waveguide coupler and the fourth waveguide coupler comprise a respective adiabatic taper within the second coupling region.

[0026] The first waveguide coupler comprises a first adiabatic taper and a second adiabatic taper within the first coupling region, wherein the first adiabatic taper has a first cross section and the second adiabatic taper has a second cross section that is different than the first cross section.

[0027] In another aspect, in general, an apparatus comprises: a first chip comprising (1) a first surface in proximity to a first set of two or more waveguide couplers, and (2) a set of two or more optical phase shifters coupled to the first set of two or more waveguide couplers; a second chip comprising (1) a second surface in proximity to a second set of two or more waveguide couplers, and (2) a set of two or more optical antennas coupled to the second set of two or more waveguide couplers; and a controller to control respective phase shifts of the optical phase shifters to transmit and/or receive an optical beam using an optical phased array comprising the two or more optical antennas; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, and a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region.

[0028] Aspects can include one or more of the following features.

[0029] The second chip further comprises (3) at least one optical distribution network coupled to a third set of two or more waveguide couplers.

[0030] The first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which each waveguide coupler of the first set of two or more waveguide couplers is in proximity to a different respective waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the each waveguide coupler of the first set of two or more waveguide couplers and the different respective waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the each waveguide coupler and the different respective waveguide coupler that are overlapped within a respective coupling region.

[0031] The interface between the first surface and the second surface is configured to suppress crosstalk between adjacent coupling regions.

[0032] Adjacent coupling regions are separated from each other by a separation distance that is larger than a maximum length of each of the coupling regions of the adjacent coupling regions.

[0033] Each coupling region is offset from an adjacent coupling region along a respective propagation axis associated with an optical wave traveling through the coupling region.

[0034] Aspects can have one or more of the following advantages.

[0035] High-density optical chip-to-chip interfaces are provided to enable modular packaging and assembly, resulting in a performance gain and cost reduction in silicon photonic microsystems. The potential cost penalties from over-integration can be mitigated through modular packaging and assembly flows. A silicon photonic circuit constructed using an active process, yet area-dominated by passive components, can be cost-inefficient. In some implementations, a two-chip solution can allow the performance of the active photonics to be optimized separately from the passive process, yielding performance gains in both if the modular packaging and assembly flow carries a low loss penalty and is high-density. The high cost-per-area of the active subcircuit can be reduced to the area solely required, and the remainder of the low cost-per-area passive circuit can result in a net cost reduction of the microsystem.

[0036] Some implementations of modular packaging and assembly can enable the performance of photonic systems with complementary metal-oxide-semiconductor (CMOS) integration to be optimized with lower cost. CMOS process improvement is rapid, with new process nodes available from foundries every two to three years. Depending on the microsystem requirements, a CMOS process suitable for an application requiring high-speed digital computation may be a poor choice for another application requiring high-voltage control. Modular packaging and assembly can allow any optimal CMOS process to be arbitrarily integrated with silicon photonics, and in such a way that reduced CMOS area is used to avoid detrimental wasted area on expensive silicon.

[0037] Other features and advantages will become apparent from the following description, and from the figures and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0039] FIGS. 1-5 are schematic diagrams of example assembled systems.

[0040] FIGS. 6A-6J are schematic diagrams of stages of an example fabrication process flow.

[0041] FIGS. 7A-9B are schematic diagrams of example couplers.

[0042] FIGS. 10A-10B, and 11 are schematic diagrams of example coupler arrays.

[0043] FIGS. 12A-12B are schematic diagrams of an example coupler comprising gratings.

[0044] FIG. 13 is a schematic diagram of an example coupler array.

[0045] FIG. 14A is a schematic diagram of an example device.

[0046] FIGS. 14B-14M are schematic diagrams of stages of an example fabrication process flow.

DETAILED DESCRIPTION

[0047] Included herein are a variety of examples of techniques for modular packaging and assembly, including an overall processing flow that can be used to form the assembled chips. Some implementations integrate passive and active photonic chips with CMOS chips in a way that enables high-density optical coupling between chips and cost-efficient use of photonic fabrication processes. As referred to herein, a “chip” may refer to an entire wafer or a portion of a wafer, such

as a die formed from a wafer that has been diced into multiple dies (also referred to as chiplets). These chips utilize photonic interconnects that allow light to be routed between the active and passive photonic-coupled chips. A variety of examples of these high-density photonic chip-to-chip interconnects are described.

[0048] FIG. 1 shows an example device **100** comprising a modular packaging scheme for an optical phased array (OPA). The device **100** comprises an active PIC **102** bonded to a passive PIC **104**. In some examples, the active PIC **102** can be configured to transmit or receive coherent optical signals. The active PIC **102** comprises a first surface in proximity to a set of waveguide couplers (not shown) that are coupled to optical phase shifters **106**. The passive PIC **104** comprises a second surface in proximity to a set of waveguide couplers (not shown) that are coupled to an array of optical grating antennas **108**. The active PIC **102** and the passive PIC **104** are optically coupled to each other by an optical coupler region **110** between the first surface of active PIC **102** and the second surface of the passive PIC **104**. The optical coupler region **110** comprises the waveguide couplers of the active PIC **102** and the waveguide couplers of passive PIC **104**. In some implementations, these the waveguide couplers can be referred to as optical interconnects and can be high-density such that the optical interconnects comprise a pitch of approximately 1 μm .

[0049] In some examples, modular design can offer several advantages in chip production by enabling the use of different process nodes for the passive and active PICs. High-temperature annealing, which can be incompatible with active chip doping, can be applied to passive chips, thereby reducing losses in waveguides and minimizing phase defects that impair beam quality. Active chips, which can be more expensive per unit area, can benefit from cost savings by minimizing their usage in routing areas and focusing them on critical functionalities. Additionally, limitations in reticle size can be mitigated, allowing more area to be dedicated to emitters, which can enhance beam directivity. The modular approach can support reconfigurability and can enable the same active photonic integrated circuit (PIC) to accommodate interchangeable grating antenna designs. Subcomponents can also be pre-yielded before bonding, which can improve overall manufacturing efficiency. Evanescent coupling with mechanical references can relax vertical alignment constraints, and high-density optical coupling can help reduce the formation of grating lobes, which can further enhance system performance.

[0050] FIG. 2 depicts an example device **200** comprising a modular packaging scheme. The device **200** comprises an active PIC **202** that is bonded to two passive PICs **204**, **206**. In some implementations, this configuration can enable the transmission or reception of coherent optical signals. The passive PIC **204** includes an optical distribution network **208** that is configured to split a coherent optical input into multiple spatially separated channels. The active PIC **202** comprises an array of optical phase shifters **210**, while the passive PIC **206** comprises an array of optical grating antennas **212**. Optical coupler region **214** interconnects the active PIC **202** and the passive PIC **204** while optical coupler region **216** interconnects the active PIC **202** and the passive PIC **206**. In some examples, the optical coupler regions **214**, **216** can each comprise high-density optical couplers with approximately 1 μm pitch providing optical interconnects between these components. The active PIC **202** is bonded to a CMOS chip **218**, which can be configured to manage the array of optical phase shifters **210**. In some examples, the array of optical phase shifters **210** can be configured as a high-density array of phase shifters. The CMOS chip **218** comprises through-silicon vias (TSVs, not shown) to route electrical power and control signals to a ball grid array (BGA) **220**. In some examples, routing electrical power and control signals can place components of a device in electrical communication with each other. The passive PIC **204** and the passive PIC **206** are each mounted to respective supports **222**, **224**. In some examples, the supports **222**, **224** can provide mechanical stability. The entire assembly is integrated with a printed circuit board (PCB) or substrate **226**.

[0051] FIG. 3 shows a cross-section view of a device **300** that can be used as a LiDAR system. In some implementations, the device **300** can be configured using modular packaging and assembly

flow. The device **300** comprises (1) active photonic components (such as phase shifters and/or photodiodes) on an active chip **302** and (2) a large region acting as one or more transmit/receive apertures that include only passive components (e.g., optical antennas, and/or optical splitter trees) on a passive chip **304**. In some examples, the active chip **302** can be referred to as an “active photonic chip” and the passive chip **304** can be referred to as a “passive photonic chip.” A variety of other types of active and passive photonic-coupled chips (including processors or other active electronic chips that use photonic signals that are converted to/from electronic signals when coupling to the electronic circuitry) can be coupled together using the photonic interface. Fabricating both the active and passive components using a single process can be inefficient due to the large area occupied by the transmit/receive apertures which do not make use of the active components available in the process. The packaging process illustrated in FIG. 3 can address this inefficiency by making use of a separately fabricated passive chip **304** and active chip **302**. For example, the active chip **302** can comprise a photonic wafer that can be fabricated using an active process which implements the active components used by the system. The active chip **302** comprises layers such as metals, doped silicon, and Germanium. The active chip **302** also comprises through oxide vias (TOV) which allow for electrical connections to be made to a CMOS chiplet **306**. The active chip **302** is then bonded on the wafer scale to the passive chip **304**. The CMOS chiplet **306** comprises components such as DACs and ADCs that interface with the active silicon photonic components. The CMOS chiplet **306** also comprises through silicon vias (TSV) that can allow electrical connections to pass through the CMOS chiplet **306**. In some examples, a hybrid active-photonic-CMOS wafer can be formed by bonding an active chip to a CMOS wafer and then dicing the wafer to yield chiplet containing an active chip bonded to a CMOS chip. In some examples, the passive chip **304** can be formed from a separate passive wafer and can comprise passive photonic components required by the system, such as transmit and receive apertures for an optical phased array LiDAR system. The passive wafer can undergo processing, including replacement of the handle wafer with a glass wafer (discussed below), and can then be diced into chips, and bonded on the chip scale to the surface of the active photonic chip. In some examples, the bond surface can be confined to a region and does not need to cover the full surface of the chips. As discussed in more detail below, optical couplers can allow light to be passed between the active chip **302** and the passive chip **304**. As shown in FIG. 3, the CMOS chiplet **306** can be attached to a substrate, such as a PCB or ceramic substrate. The TSVs on the CMOS chiplet **306** can be electrically connected to pads on the substrate (e.g., using solder balls). This connection can allow components on the CMOS chiplet **306** to interface electrically with components on the substrate. Further, since the TSVs connect up to the photonic components, they can also provide a path for the photonic components to connect electrically to the PCB/substrate. The TOVs on the active chip **302** can allow electrical connection to be made to the CMOS chiplet **306** and also up to the passive photonic components if needed. To allow for the integration of gain media that can be used for lasers and optical amplifiers, a trench can be etched through the passive chip **304** and into the active chip **302**, the trench can be metallized, and III/V materials can be flipped into the trench, as illustrated in FIG. 3. In some examples, the trench can be etched only through an active chip if the passive chip is not present in the region where the trench is being etched. More details about the process flow used to implement the above packaging process are discussed below.

[0052] FIG. 4 shows a cross-section view of a device **400** that is produced using a packaging process similar to the one described above. The device **400** comprises an active photonic chip **402**, a passive photonic chip **404**, and a CMOS chiplet **406**. The CMOS chiplet **406** of the device **400** does not include TSVs. Instead, metal layers are included in the passive photonic chip **404** so that electrical connections from the CMOS chiplet **406** and active photonic chip **402** can be routed up to the metals in the passive photonic chip **404** using the TOVs for connections originating from the CMOS chiplet **406**. The connections can then be routed to pads located on the passive photonic chip **404** which are then electrically connected to a PCB/ceramic substrate using, for example, solder

balls. In this process, the CMOS chiplet **406** is attached to the substrate using an adhesive such as an epoxy.

[0053] FIG. **5** shows a cross-section view of an example device **500** formed from a modular packaging process applied to enable high-density chip-to-chip optical interconnects that interface with a processor. The device **500** comprises an active photonic chip **502**, a passive photonic chip **504**, and a CMOS chiplet **506**. As in the processes discussed earlier, a CMOS wafer and active photonic wafer can be bonded on the wafer scale and then diced into hybrid active-photonic-CMOS chip comprising the active photonic chip **502** and the CMOS chiplet **506**. This hybrid active-photonic-CMOS chip can then be bonded on the chip scale to a passive photonic chip **504** fabricated with a separate process. In this example, the passive photonic chip **504** contains through glass vias (TGV) that allow electrical connections to be made to a processor residing above the passive photonic chip **504**. The active photonic chip **502** comprises TOVs which allow electrical connections to pass through the active photonics chip, and the CMOS chiplet **506** comprises TSVs that allow connections to pass through the CMOS chiplet **506** to a PCB/ceramic substrate. Using the TGVs and the TOVs, the processor can transfer data electrically to the CMOS chiplet **506**, which can then drive the active components on the active photonic chip **502**. By driving the components on the active photonic chip **502**, e.g., modulators connected to laser source(s), data can be encoded in an optical signal that can then be routed from the active photonic chip **502** up to the passive photonic chip **504** using optical couplers that are described in more detail below. Light can then be routed through the passive photonic chip **504** to a region that is bonded to a second active-photonic-CMOS hybrid chip (not shown), which can be possibly connected to another processor. The light can be coupled to the second hybrid chip using the optical couplers that are discussed in more detail below.

[0054] FIGS. **6A-6J** show an example process for assembling devices. FIG. **6A** shows a step **600A** comprising configuring a CMOS device **602** from a portion of a wafer and a silicon photonic device **604** from a portion of a wafer. In some examples, the silicon photonic device **604** can be an active photonic device or chip. The CMOS device **602** and the silicon photonics device **604** each comprise respective components identified by the key to the right of FIG. **6A**. In some examples, the silicon photonics device **604** can be fabricated using an active process. During step **600B**, as shown in FIG. **6B**, the silicon photonics device **604** is flipped onto the CMOS device **602**. In some examples, this flip-chip step can be performed at the wafer scale such that the wafer comprising the CMOS device **602** and the wafer comprising the silicon photonics device **604** are bonded on the wafer scale. The bonding process includes a bond of the wafer oxide surfaces and a bond of the metal pads to allow electrical connection between the CMOS device **602** and the silicon photonic device **604**. After bonding the wafers, the silicon handle substrate is removed from the silicon photonic device **604**, as illustrated in step **600C** depicted in FIG. **6C**. After removal of the handle wafer, through oxide vias (TOVs) are added to allow electrical connections to pass through the photonic device layer down to the CMOS device **602**, as shown step **600D** depicted in FIG. **6D**. After adding the TOVs, the wafer can be diced into individual chips. A glass interposer **606** is constructed by bonding a glass wafer to a passive silicon photonic wafer and then debonding the silicon handle wafer, as shown in FIG. **6E**. The glass interposer **606** is then diced into individual chips and bonded to the silicon photonic device **604**, as shown in step **600F** depicted in FIG. **6F**. The resulting device **608**, shown in step **600G** depicted in FIG. **6G** comprises the CMOS device **602**, the silicon photonic device **604**, and the glass interposer **606**. In other words, the device **608** comprises an active photonic chip, i.e., the silicon photonic device **604**, and a passive photonic chip, i.e., the glass interposer **606**. To allow gain media to be integrated into the chip, a trench **610** can be etched down into the device **608** through the glass interposer **606** and the silicon photonic device **604**, as shown by step **600H** depicted in FIG. **6H**. A chip **612** containing III/V materials, for example, with quantum wells implemented on an InP platform, can then be flipped into the trench **610**, yielding the device **608** shown in FIG. **6I**. This assembly process can be used to couple

multiple hybrid active-photonic-CMOS chips into a larger chip by using the passive photonic chip to route light between the active chips. This coupling is illustrated in FIG. 6J, which shows a first hybrid active-photonic-CMOS chip **620** and a second hybrid-active-photonic CMOS chip **622** coupled by a single passive photonic chip **624**. While coupling between two chips is shown in FIG. 6I, the methods described here can be used to couple an arbitrary number of such chips, with dense optical routing between the chips enabled by one or more passive photonic chips. In some examples, a CMOS wafer can also comprise TOVs.

[0055] The modular packaging process described above makes use of photonic chips which are stacked vertically and bonded over an interface through which light can be transferred between the chips. For example, an active photonic circuit which includes an array of optical phase shifters can be bonded to a region of a passive photonic circuit which contains an array of antennas so that, by combining the functionality of both chips, the device can operate as an optical phased array where the beam emitted by the antennas on the passive chip is steered by the phase shifters residing on the active chip. As another example, if a system requires a photonic integrated circuit which contains active components fabricated with a process that is incompatible with low-loss waveguides, an active photonic chip could be interfaced to a passive photonic chip where the waveguide loss can be kept low. To reduce losses, light from the active chip could then be coupled into the passive chip, routed over a long distance with minimal loss, and then coupled back into the active chip. Passing light across the interface between chips requires the use of an optical coupler. Such a coupler can be implemented using adiabatic couplers or using gratings (discussed in more detail below). As discussed below, these couplers can be arrayed to enable high-density chip-to-chip photonic interconnects.

[0056] FIG. 7A shows a perspective view of an example adiabatic coupler **700** that can be used to couple light vertically across the interface between two photonic chips. FIG. 7B depicts a top view of the example adiabatic coupler **700**. The adiabatic coupler **700** comprises a first waveguide coupler **702** and a second waveguide coupler **704**. In some implementations, a first chip (not shown) can comprise the first waveguide coupler **702** and a second chip (not shown) can comprise the second waveguide coupler **704**. As shown in FIG. 7A, the first waveguide coupler **702** and the second waveguide coupler **704** are stacked in the z-direction, so that the first waveguide coupler **702** is positioned at a larger z-coordinate compared to the second waveguide coupler **704**. The first waveguide coupler **702** is in proximity to the second waveguide coupler **704** within a distance that supports evanescent optical coupling between the first waveguide coupler **702** and the second waveguide coupler **704** that are overlapped within a coupling region **706**. In some implementations, the interface between the first chip and the second chip can form a plane parallel to the x-y plane. In some examples, the interface between the chips does not need to cover the entire surface of either chip but can be restricted to a region or portion of each of the chips. As illustrated in FIGS. 7A-7B, the widths of the first waveguide coupler **702** and the second waveguide coupler **704** are adiabatically tapered along the length of the waveguide couplers in the coupling region **706**. This adiabatic tapering allows the mode associated with one waveguide coupler to be adiabatically converted into the mode associated with the other waveguide coupler, allowing light to be transferred from one waveguide coupler to the other. In some examples, each waveguide coupler can be at the end of a corresponding waveguide such that light from a waveguide can propagate to the first waveguide coupler **702**, be coupled into the second waveguide coupler **704**, and then propagate into another waveguide from the second waveguide coupler **704**.

[0057] To implement an adiabatic coupler between two photonic chips, the chips can be aligned so that the tapered waveguides on the two chips are in close proximity. However, imperfections in the packaging process can result in the chips being misaligned and, as a result, the coupling efficiency can decrease. To decrease the sensitivity of the coupler to misalignment, the taper can be designed so that the mode expands significantly as it propagates through the taper. In such a design, distributing the mode over a larger spatial region (i.e., such that the mode has a larger mode

diameter) can enable the device to be less sensitive to misalignment between a tapered waveguide in a first chip and a tapered waveguide in a second chip.

[0058] FIG. 8A shows a perspective view of an example adiabatic coupler **800** and FIG. 8B shows a top view of the adiabatic coupler **800**. The adiabatic coupler **800** comprises a first waveguide coupler **802** and a second waveguide coupler **804**. The first waveguide coupler **802** is in proximity to the second waveguide coupler **804** within a distance that supports evanescent optical coupling between the first waveguide coupler **802** and the second waveguide coupler **804** that are overlapped within a coupling region **806**. In this design, the first waveguide coupler **802** and the second waveguide coupler **804** each comprise portions in the coupling region **806** where the width of each waveguide coupler **802**, **804** is small so that the mode in the respective portion has a large mode diameter. As a result of these portions, the adiabatic coupler **800** can tolerate larger misalignments between the two waveguides in the y and z directions.

[0059] FIGS. 9A-9B shows another example adiabatic coupler **900** that can similarly achieve improved alignment tolerances by expanding the mode diameter in the region with the adiabatic taper. FIG. 9A depicts a perspective view of the adiabatic coupler **900** while FIG. 9B depicts a top view of the adiabatic coupler **900**. The adiabatic coupler **900** comprises a first waveguide coupler **902** and a second waveguide coupler **904**. In some examples, the first waveguide coupler **902** can be in a first chip (not shown) and the second waveguide coupler **904** can be in a second chip (not shown) to facilitate coupling between the first chip and the second chip. The first waveguide coupler **902** is in proximity to the second waveguide coupler **904** within a distance that supports evanescent optical coupling between the first waveguide coupler **902** and the second waveguide coupler **904** that are overlapped within a coupling region **906**. In this example, the first waveguide coupler **902** is constructed using two different layer heights shown by layers **908**, **910** that are formed from a core guiding material. For example, the core guiding material could be constructed from a combination of a **220** nm height silicon layer combined with a partial ridge etch that produces a region with a reduced silicon height layer. In some examples, the **220** nm silicon layer can be referred to as a full-height layer **908** while the reduced silicon height layer can be referred to as a partial-height layer **910**. With reference to FIGS. 9A-9B, the word height refers to the waveguide dimension measured along the z direction. While silicon is used as an example material here, other materials can also be used to form the waveguide couplers. The adiabatic taper of the first waveguide coupler **902** comprises two stages. First, the width (measured in the y-direction) of the full-height layer **908** is tapered to transfer the mode into the partial-height layer **910**. Then the width, as measured in the y-direction, of the partial-height layer **910** is tapered down to further expand the mode. In other words, the first waveguide coupler **902** comprises a first adiabatic taper and a second adiabatic taper within the coupling region **906**, where the first adiabatic taper has a first cross section and the second adiabatic taper has a second cross section that is different than the first cross section. By transferring the mode into a narrow-width region composed of partial-height layer **910**, the mode can have a larger mode diameter than it would if the taper were constructed purely using the full-height layer **908**. The second waveguide coupler **904** also tapers to a region where the mode has a large diameter. In FIGS. 9A-9B, the taper of the second waveguide coupler **904** is shown as being constructed from a single-height core layer. This single-height design could be used, for instance, if the material used to construct the second waveguide coupler **904** is a low index contrast material, such as silicon nitride embedded in glass, which may make the use of a partial-height region unnecessary. However, a two-height design such as the one used for the taper of the first waveguide coupler **902** could also be used for the second waveguide coupler **904**. In that case, the waveguide structures can be inverted so that the wider portion of the full-height silicon is on the top instead of the bottom, or the waveguide structures can have the same orientation with the wider portion of the full-height silicon on the bottom and the narrower portion of the full-height silicon on the top. In either orientation, the two partial-height regions can be adjacent to each other such that they overlap each other.

[0060] By fabricating an array of the type of adiabatic couplers described above, many independent optical channels can be coupled between photonic chips, allowing for implementation of dense optical chip-to-chip interconnects. An example of this type of configuration is illustrated in FIG. **10A**, which shows a top view (in the xy plane) of an adiabatic coupler array **1000A** where the individual couplers in the array utilize the design previously shown in FIGS. **8A-8B**. The adiabatic coupler array **1000A** comprises a first plurality of waveguide couplers **1002A-1002N** and a second plurality of waveguide couplers **1004A-1004N**. In some examples, the first plurality of waveguide couplers **1002A-1002N** can be in a first photonic chip (not shown) and the second plurality of waveguide couplers **1004A-1004N** can be in a second photonic chip (not shown). Each waveguide coupler of the first plurality of waveguide couplers **1002A-1002N** is overlapped with a respective waveguide coupler of the second plurality of waveguide couplers **1004A-1004N** in a respective coupling region **1006A-1006N**. As shown in FIG. **10A**, each waveguide coupler of the first plurality of waveguide couplers **1002A-1002N** and each waveguide coupler of the second plurality of waveguide couplers **1004A-1004N** comprise respective adiabatic tapers within the respective coupling regions **1006A-1006N**. In some implementations, other coupler designs can also be used for the individual couplers in the array. As before, the two photonic chips are assumed to be stacked in the z-direction (out of the page in FIG. **10A**), so that the first chip resides at a larger z-coordinate compared to the second chip and the interface between the two chips forms a planar region parallel to the x-y plane.

[0061] In some examples, each waveguide coupler of the first plurality of waveguide couplers **1002A-1002N** can be at an end of a corresponding waveguide on the first photonic chip (not shown). In some examples, each of these corresponding waveguides can be arranged to have respective propagation axes that are all parallel to a common axis in this region of the first photonic chip. For instance, each of the corresponding waveguides can have respective propagation axes that are all parallel to the x-axis of FIG. **10A**. In some examples, each waveguide coupler of the first plurality of waveguide couplers **1002A-1002N** can be substantially coplanar with a first plane that is parallel to this common axis, i.e., the x-axis. The first plane could be, for instance, a plane that is parallel to the xy plane. Likewise, each waveguide coupler of the second plurality of waveguide couplers **1004A-1004N** can be at an end of a corresponding waveguide on the second photonic chip (not shown). Similarly, each of the corresponding waveguides can have respective propagation axes that are all parallel to a common axis in this region of the second photonic chip. In some examples, each waveguide coupler of the second plurality of waveguide couplers **1004A-1004N** can be substantially coplanar with a second plane that is parallel to each of the common axis, i.e., the x-axis, and the first plane. The second plane could be, for instance, a plane that is parallel to the xy plane.

[0062] As shown in FIG. **10A**, a portion of each waveguide coupler of the first plurality of waveguide couplers **1002A-1002N** that is contained within a coupling region **1006A-1006N** can be contained within a third plane that is perpendicular to the first plane and perpendicular to the common axis, i.e., the x-axis. In this example, the third plane could be a plane that is parallel to the yz plane. Further, a portion of each waveguide coupler of the second plurality of waveguide couplers **1004A-1004N** that is contained within a coupling region **1006A-1006N** can be contained within a fourth plane that is perpendicular to the first plane and perpendicular to the common axis. In this example, the fourth plane could be a plane that is parallel to the yz plane such that the third plane and the fourth plane are distributed along the common axis.

[0063] By way of example, referring to FIG. **10A**, the waveguide coupler **1002B** can be separated from the waveguide coupler **1002A** and the waveguide coupler **1002C** by a separation distance along the y-axis. Separating the waveguide couplers in this way can also separate the associated coupling regions. By way of example, the coupling region **1006B** is separated from the coupling region **1006A** by separation distance **1008A** and is separated from the coupling region **1006C** by separation distance **1008B**. This separation distance along the y-axis can be referred to as

waveguide pitch.

[0064] In some implementations, configuring waveguide couplers and coupling regions can be associated with reducing crosstalk between optical waves propagating in each of the waveguide couplers and/or coupling regions. Some configurations can comprise separating adjacent waveguide couplers or coupling regions by a separation distance that is based on a maximum length associated with the coupling region. For example, a maximum length of a coupling region can be length in which an optical wave transitions from the first waveguide coupler of the coupling region into the second waveguide coupler of the coupling region. In some examples, each adjacent coupling region can be separated by a separation distance that is larger than a maximum length of each of the coupling regions in the adjacent coupling regions. FIG. 10B depicts an example coupler array **1000B** comprising a larger pitch between adjacent waveguide couplers and waveguide coupling regions than the example coupler array depicted in FIG. 10A. The coupler array **1000B** comprises a first plurality of waveguide couplers **1052A-1052C** and a second plurality of waveguide couplers **1054A-1054C** overlapped in respective coupling regions **1056A-1056C**. By way of example, the coupling region **1056B** is separated from coupling region **1056A** by a separation distance **1058A** and is separated from coupling region **1056C** by a separation distance **1058B**. In some implementations, this larger pitch can be associated with reduced crosstalk between adjacent coupling regions.

[0065] Referring back to FIG. 10A, a coupler array comprising couplers spaced at a small pitch, also referred to as the separation distance between adjacent couplers, can be desirable to couple light between chips. For example, an optical phased array comprising an array of phase shifters coupled to an array of antenna elements, and implementing smaller antenna/phase-shifter pitches can enable the optical phased array to operate over a larger angular beam steering range. If the phase shifters and antennas are located on separate chips and coupled using the type of adiabatic coupler array discussed above, a coupler array with a small pitch that matches the pitch of the antennas/phase-shifters can be desirable to optimize operating capabilities. As another example, if data is being optically transmitted between the chips, a coupler array with a small pitch can enable dense data transmission from chip to chip, i.e., many channels within a small spatial region.

[0066] In some implementations, as the pitch of the couplers is decreased, undesirable crosstalk between neighboring couplers can be introduced. To mitigate this crosstalk, neighboring couplers can be staggered, as illustrated by the example coupler array **1100** shown in FIG. 11. The coupler array **1100** comprises a first plurality of waveguide couplers **1102A-1102N** and a second plurality of waveguide couplers **1104A-1104N**. Each waveguide coupler of the first plurality of waveguide couplers **1102A-1102N** is overlapped with a respective waveguide coupler of the second plurality of waveguide couplers **1104A-1104N** in a respective coupling region **1106A-1106N**. As shown in FIG. 11, adjacent coupling regions **1106A-1106N** are displaced relative to each other not only along the y-axis, but also along the axis of the waveguide couplers (the x-axis) so that the tapered regions of the coupling regions **1106A-1106N** do not overlap along the x-axis (as they do in the coupler arrays **1000A** and **1000B**). By way of example, the coupling region **1106B** can be separated by a diagonal separation distance **1108A** from the coupling region **1106A** and by a diagonal separation distance **1108B** from the coupling region **1106C**. This enables a smaller pitch without significant crosstalk. As an example, when the mode is expanded in the tapered region of the first waveguide coupler **1102B** in the coupling region **1106B**, it can be phase mismatched from the neighboring coupling regions **1106A**, **1106C** due to the fact that the neighboring waveguide couplers **1102A**, **1102C** and **1104A**, **1104C** each have a different respective cross sections near the coupling region **1106B**. This difference in cross sections can result in a mismatch in propagation constants of the modes of the coupling region **1106B**. Because the modes are phase mismatched, the crosstalk between adjacent coupling regions **1106A**, **1106C** can be reduced.

[0067] The example coupler array **1100** depicted in FIG. 11 is configured to suppress crosstalk between adjacent coupling regions. In some examples, the separation distances between adjacent

couplers can be described relative to a common axis associated with optical waves propagating in the waveguide couplers or coupling regions. By way of example, each of the waveguide couplers **1102A-1102N** can be parallel to a common axis, in this case the x-axis. Each waveguide coupler **1102A-1102N** is offset relative to adjacent waveguide couplers along the common axis such that the separation distances **1108A, 1108C** each form a nonzero angle relative to the x-axis and the y-axis.

[0068] Gratings can also be used to couple light across the interface between the stacked chips. A perspective view of an example coupler **1200** is illustrated in FIG. **12A** and a side view of the coupler **1200** is shown in FIG. **12B**. The coupler **1200** comprises a first waveguide coupler **1202** in a first chip (not shown) and a second waveguide coupler **1204** in a second chip (not shown) with propagating light shown by arrows. The first waveguide coupler **1202** and the second waveguide coupler **1204** are overlapped in coupling region **1206**. In this design, the first waveguide coupler **1202** comprises a first grating **1208** and the second waveguide coupler **1204** comprises a second grating **1210**. Light traveling through the first waveguide coupler **1202** is emitted by the first grating **1208** and then coupled into the second waveguide coupler **1204** via the second grating **1210**. While the gratings illustrated in FIGS. **12A-12B** are shown with a single layer of perturbations, multiple layers of perturbations, residing at different z coordinates, can be used to make the grating emission directional e.g., so that the first grating **1208** emits primarily downward in the negative z direction, and the second grating **1210** primarily receives light that is incident from above.

[0069] Grating couplers can also be used to couple light between arrays of waveguides on two chips. A perspective view of an example configuration of a coupling array **1300** is illustrated in FIG. **13**. The coupling array **1300** comprises a first plurality of waveguide couplers **1302A-1302N** and a second plurality of waveguide couplers **1304A-1304N**, each arranged on a respective first chip and second chip (not shown) along the y-direction. Each waveguide coupler of the first plurality of waveguide couplers **1302A-1302N** comprises a respective grating coupler **1308A-1308N** and each waveguide coupler of the second plurality of waveguide couplers **1304A-1304N** comprises a respective grating coupler **1310A-1310N**. Each grating coupler **1308A-1308N** couples light into a respective grating coupler **1310A-1310N**. This type of configuration could be useful, for example, for optical phased arrays where a large number of optical phase shifters reside on a first chip configured as an active chip, a large array of antennas reside on a second chip configured as a cheaper passive chip, and grating couplers allow light to be coupled across the interface between the two chips. One benefit of this type of configuration is that it can allow the grating antennas or grating couplers, which can occupy a large chip area, to be implemented using a cheaper passive process, which can reduce the overall system cost.

[0070] FIG. **14A** shows a cross-section view of an example device **1400A** comprising an active photonics chip **1402**, a passive photonics chip **1404**, an optical source **1406**, i.e., an integrated laser or an optical amplifier, and a CMOS chiplet **1408**. In this example, the CMOS chiplet **1408** includes copper pillars for flip-chip attachment to the active photonics chip **1402**. The optical source **1406** is heterogeneously integrated onto the passive photonics chip **1404** and optically coupled to patterned waveguides. Optical waves are produced by the optical source and then coupled into the patterned waveguides. The optical waves **1409** from the patterned waveguides of the passive photonics chip **1404** are evanescently coupled into the active photonics chip **1402**. Active components in the active photonics chip **1402**, i.e., modulators or phase shifters, can interact with the optical waves. Following this interaction, optical waves **1410** are evanescently coupled back into the passive photonics chip **1404**, where emission structures, i.e., grating couplers or optical antennas, emit the optical wave into free space. While the optical source **1406** can be powered by external circuitry (not shown), at least a portion of the passive photonics chip **1404** comprises passive components, such as the emission structures from which light is emitted.

[0071] As shown in FIG. **14A**, optical waves **1410** are evanescently coupled from the active

photonics chip **1402** into the passive photonics chip **1404** before they are emitted. In other words, the active photonics chip **1402** and the passive photonics chip **1404** are optical coupled by a first interface between a surface of the active photonics chip **1402** and a surface of the passive photonics chip **1404**. These optical waves **1410** can be evanescently coupled in this first interface via a first set of two or more waveguide couplers arranged near the surface of the active photonics chip **1402** and a second set of two or more waveguide couplers arranged near the surface of the passive photonics chip **1404**, such as the coupler arrays shown in FIGS. **10A-10C**, FIG. **11**, or FIG. **13**. The optical waves **1409** are evanescently coupled from the passive photonics chip **1404** to the active photonics chip **1402**. In other words, the active photonics chip **1402** and the passive photonics chip **1404** are optical coupled by a second interface between the surface of the active photonics chip **1402** and the surface of the passive photonics chip **1404**. The optical waves **1409** can be evanescently coupled in this second interface via a third set of one or more waveguide couplers arranged near the surface of the active photonics chip **1402** and a fourth set of one or more waveguide couplers arranged near the surface of the passive photonics chip **1404**, such as the waveguide couplers shown in FIGS. **7A-7B**, FIGS. **8A-8B**, or FIGS. **12A-12B**. In some examples, components that are configured to transmit and couple optical waves in this manner, such as waveguides or one or more waveguide couplers of a set of waveguide couplers, can be described as being in optical communication with the optical source **1406**.

[0072] In some examples, one or more optical splitters (not shown) can be included in the active photonics chip **1402** such that the optical waves **1409** that are evanescently coupled between the third set of one or more waveguide couplers and the fourth set of one or more waveguide couplers can be distributed into multiple waveguides for interaction with active components. Following interaction, the optical waves **1410** can then be evanescently coupled between the first set of two or more waveguide couplers and the second set of two or more waveguide couplers.

[0073] In some examples, one or more optical splitters (not shown) can be included in the passive photonics chip **1404** to distribute the optical waves **1409** into multiple waveguides. Following the one or more optical splitters, the optical waves **1409** can be evanescently coupled between a third set of two or more waveguide couplers and a fourth set of two or more waveguide couplers, such as the coupler arrays shown in FIGS. **10A-10C**, FIG. **11**, or FIG. **13**.

[0074] FIGS. **14B-14M** depict steps **1400B-1400M** associated with assembling the components into device **1400A**. During step **1400B** shown in FIG. **14B**, a passive photonics chip **1404** is formed from fabricated silicon-on-insulator. The passive photonics chip **1404** comprises a trench **1411** with metallization/patterned materials to facilitate integration of the optical source **1406** (not shown). The passive photonics chip **1404** further comprises patterned silicon-on-insulator and additional materials of sufficient optical transparency to facilitate routing of optical signals. Optical antennas **1412** are also included for transmitting and receiving optical signals with at least one layer close to surface of oxide capable of supporting evanescent coupling.

[0075] During step **1400C** as shown in FIG. **14C**, an optical source **1406** is configured. As previously mentioned, some optical sources can comprise an integrated laser or an optical amplifier. Some optical sources comprise a material that is different than a material from which a device is formed such that the optical source is heterogeneous. Some integrated lasers or optical amplifiers can comprise a laser gain material. In this example, the optical source **1406** comprises indium phosphide.

[0076] During step **1400D** shown in FIG. **14D**, a CMOS chiplet **1408** is configured. The CMOS chiplet **1408** comprises copper pillars such that the CMOS chiplet **1408** can be attached to the active photonics chip **1402**, as shown in FIG. **14A**.

[0077] An example electronic interposer chip **1414** is configured in step **1400E** shown in FIG. **14E**. The electronic interposer chip **1414** comprises wettable bond pads for copper pillar CMOS attach arranged on a top side of the electronic interposer chip **1414**. The electronic interposer chip **1414** further comprises metal and oxide interfaces for hybrid bonding arranged on a bottom surface of

the electronic interposer chip **1414**. Through-silicon vias support electrical routing between the bond pads arranged on the top surface and the hybrid bonding sites arranged on the bottom surface of the electronic interposer chip **1414**.

[0078] An example photonics chip **1416** is configured in step **1400F** shown in FIG. **14F**. The photonics chip **1416** is formed from silicon-on-insulator and comprises a thin-buried oxide to support evanescent coupling of optical waves between the photonics chip **1416** and other components (not shown). In some examples, a thin-buried oxide can comprise a thickness of $<1\text{ }\mu\text{m}$ to support efficient evanescent coupling of optical waves. The photonics chip **1416** further comprises a metal and oxide interface co-designed with the electronic interposer for hybrid bonding arranged on a top surface of the photonics chip **1416**. In some examples, the metal and oxide interface can be formed in a back end of line process such that one or more metal layers of are deposited onto patterned wafer. In some examples, the photonics chip **1416** can comprise n- and p-type doping to enable electro-optic or thermo-optic phase shifting. One or more metal layers are also included for optical routing. The photonics chip **1416** further comprises one or more via layers to enable electrical connections between metal layers and contact layers from metal to doped silicon-on-insulator.

[0079] FIG. **14G** shows step **1400G** in which the electronic interposer chip **1414** and the photonics chip **1416** arranged for wafer-scale hybrid bonding. A cross-section view of the electronic interposer chip **1414** and the photonics chip **1416** following wafer-scale hybrid bonding is shown in step **1400H** in FIG. **14H**. Following wafer-scale hybrid bonding, the backside silicon substrate handle can be removed from the photonics chip **1416**, as shown by step **1400I** shown in FIG. **14I**. After removal of the backside silicon substrate handle, dicing/segmentation of the hybrid bonded wafers can be performed to form one or more active photonics chips, as shown by step **1400J** in FIG. **14J**. FIG. **14J** depicts a plurality of active photonics chips **1402**, **1418**, **1420** including the active photonics chip **1402** shown in FIG. **14A**. Each active photonics chip **1402**, **1418**, **1420** comprises a portion of the electronic interposer chip **1414** and a portion of the photonics chip **1416**. [0080] FIG. **14K** depicts step **1400K** in which the optical source **1406** is integrated onto the passive photonics chip **1404** via the trench **1411**. Following integrating, the active photonics chip **1402** can be oxide bonded to the passive photonics chip **1404**, as shown by step **1400L** in FIG. **14L**. The CMOS chiplet **1408** can then be flip-chip attached to the active photonics chip **1402**, as shown by step **1400M** in FIG. **14M**, to form the device **1400A**.

[0081] While the disclosure has been described in connection with certain embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. An apparatus, comprising: a first chip comprising a first surface in proximity to a first set of two or more waveguide couplers; and a second chip comprising a second surface in proximity to a second set of two or more waveguide couplers; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler

that are overlapped within a second coupling region, and the first coupling region is separated from the second coupling region by a separation distance that is larger than a maximum length of the first coupling region and larger than a maximum length of the second coupling region.

2. The apparatus of claim 1, wherein each waveguide coupler in the first set of two or more waveguide couplers is at an end of a corresponding waveguide on first chip, and each waveguide coupler in the second set of two or more waveguide couplers is at an end of a corresponding waveguide on the second chip.

3. The apparatus of claim 2, wherein the corresponding waveguides on the first chip are arranged (1) to have propagation axes that are all parallel to a common axis and (2) such that corresponding waveguide couplers of waveguides that are adjacent to each other on the first chip are offset from each other along the common axis.

4. The apparatus of claim 3, wherein the first waveguide coupler and the third waveguide coupler are substantially coplanar with a first plane that is parallel to the common axis and the second waveguide coupler and the fourth waveguide coupler are substantially coplanar with a second plane that is parallel to each of the common axis and the first plane.

5. The apparatus of claim 4, wherein a portion of the first waveguide coupler and a portion of the second waveguide coupler within the first coupling region are contained within a third plane that is perpendicular to the first plane and perpendicular to the common axis, and a portion of the third waveguide coupler and a portion of the fourth waveguide coupler within the second coupling region are contained within a fourth plane that is perpendicular to the first plane and perpendicular to the common axis.

6. The apparatus of claim 5, wherein the third plane and the fourth plane are distributed along the common axis.

7. The apparatus of claim 1, wherein each of the first waveguide coupler, the second waveguide coupler, the third waveguide coupler, and the fourth waveguide coupler comprise respective grating couplers.

8. The apparatus of claim 1, wherein each waveguide coupler of the first set of two or more waveguide couplers is optically coupled to a different respective optical phase shifter.

9. The apparatus of claim 1, wherein each waveguide coupler of the second set of two or more waveguide couplers is optically coupled to a different respective optical antenna.

10. The apparatus of claim 1, wherein the first surface of the first chip is in proximity to a third set of one or more waveguide couplers and the second surface of the second chip is in proximity to a fourth set of one or more waveguide couplers.

11. The apparatus of claim 10, wherein the first chip and the second chip are optically coupled to each other by a second interface between the first surface and the second surface in which: a fifth waveguide coupler of the third set of one or more waveguide couplers is in proximity to a sixth waveguide coupler of the fourth set of one or more waveguide couplers within a distance that supports evanescent optical coupling between the fifth waveguide coupler and the sixth waveguide coupler that are overlapped within a third coupling region.

12. The apparatus of claim 11, wherein at least one waveguide coupler in the fourth set of one or more waveguide couplers is in optical communication with an optical source on the second chip.

13. An apparatus, comprising: a first chip comprising a first surface in proximity to a first set of two or more waveguide couplers; and a second chip comprising a second surface in proximity to a second set of two or more waveguide couplers; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide

coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region, the first waveguide coupler and the third waveguide coupler are each at an end of a waveguide that is coupled to an active photonic component, and the second waveguide coupler and the fourth waveguide coupler are each at an end of a waveguide that is coupled to a passive photonic component.

14. The apparatus of claim 13, wherein the active photonic component comprises an optical phase shifter.

15. The apparatus of claim 14, wherein the first chip comprises one or more through oxide vias in electrical communication with the active photonic component.

16. The apparatus of claim 14, wherein the passive photonic component comprises an optical antenna.

17. The apparatus of claim 13, wherein each of the first waveguide coupler and the second waveguide coupler comprise a respective adiabatic taper within the first coupling region, and each of the third waveguide coupler and the fourth waveguide coupler comprise a respective adiabatic taper within the second coupling region.

18. The apparatus of claim 13, wherein the first waveguide coupler comprises a first adiabatic taper and a second adiabatic taper within the first coupling region, wherein the first adiabatic taper has a first cross section and the second adiabatic taper has a second cross section that is different than the first cross section.

19. An apparatus, comprising: a first chip comprising (1) a first surface in proximity to a first set of two or more waveguide couplers, and (2) a set of two or more optical phase shifters coupled to the first set of two or more waveguide couplers; a second chip comprising (1) a second surface in proximity to a second set of two or more waveguide couplers, and (2) a set of two or more optical antennas coupled to the second set of two or more waveguide couplers; and a controller to control respective phase shifts of the optical phase shifters to transmit and/or receive an optical beam using an optical phased array comprising the two or more optical antennas; wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which: a first waveguide coupler of the first set of two or more waveguide couplers is in proximity to a second waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the first waveguide coupler and the second waveguide coupler that are overlapped within a first coupling region, and a third waveguide coupler of the first set of two or more waveguide couplers is in proximity to a fourth waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the third waveguide coupler and the fourth waveguide coupler that are overlapped within a second coupling region.

20. The apparatus of claim 19, wherein the second chip further comprises (3) at least one optical distribution network coupled to a third set of two or more waveguide couplers.

21. The apparatus of claim 19, wherein the first chip and the second chip are optically coupled to each other by an interface between the first surface and the second surface in which each waveguide coupler of the first set of two or more waveguide couplers is in proximity to a different respective waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the each waveguide coupler of the first set of two or more waveguide couplers and the different respective waveguide coupler of the second set of two or more waveguide couplers within a distance that supports evanescent optical coupling between the each waveguide coupler and the different respective waveguide coupler that are overlapped within a respective coupling region.

22. The apparatus of claim 21, wherein the interface between the first surface and the second surface is configured to suppress crosstalk between adjacent coupling regions.

23. The apparatus of claim 22, wherein adjacent coupling regions are separated from each other by

a separation distance that is larger than a maximum length of each of the coupling regions of the adjacent coupling regions.

24. The apparatus of claim 23, wherein each coupling region is offset from an adjacent coupling region along a respective propagation axis associated with an optical wave traveling through the coupling region.
